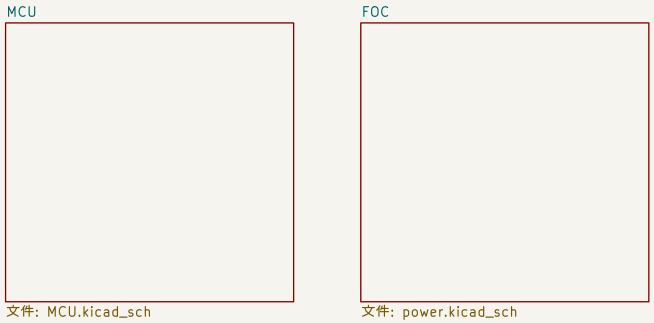
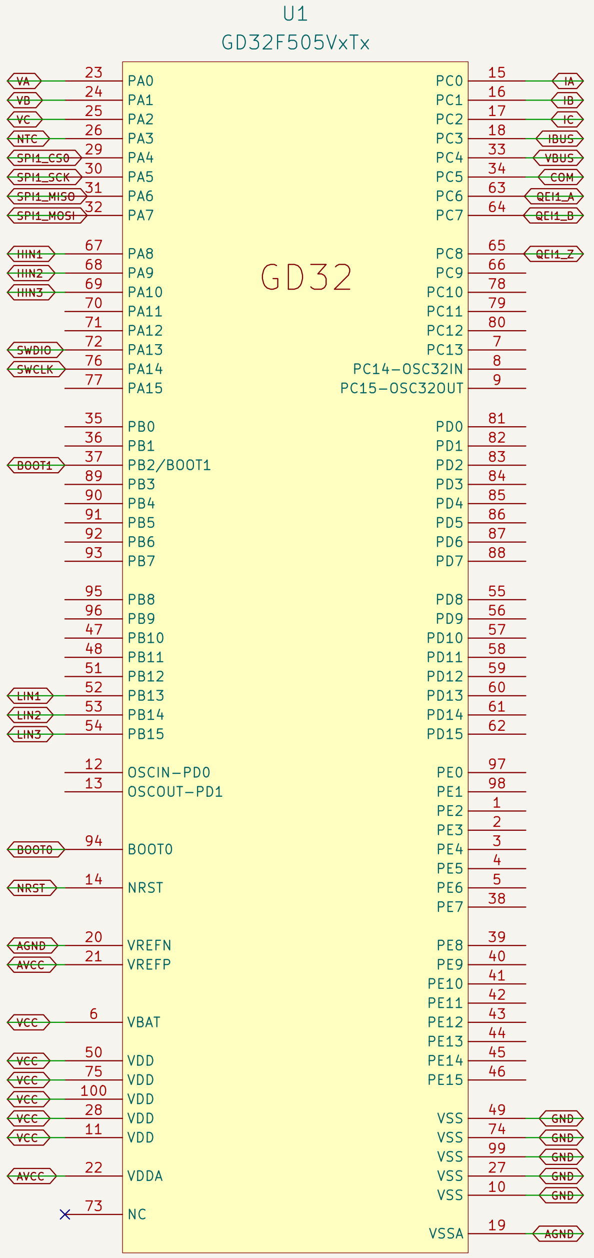
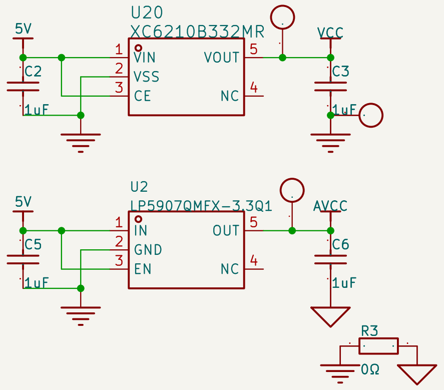


Sheet: /
File: GD32F5_FOC.kicad_sch

Title:

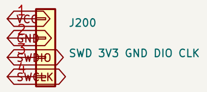
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6-87-gae6d8f40c4		Id: 1/3

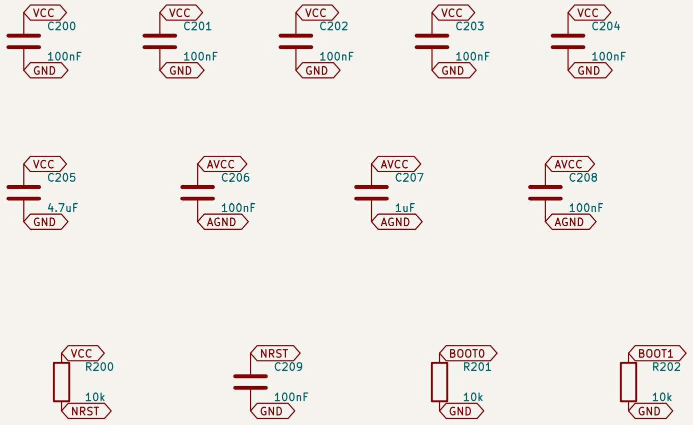




GD32F505VxTx / official GD32EB F505VxT7.xml pin mapping

HIN/LIN: TIMER0 CH0/1/2 and complementary outputs.
SPI1_* connector nets use GD32 SPI0 on PA4..PA7.
QEI1_*: TIMER2 on PC6..PC8. ADC inputs on PA0..3, PC0..5.
VCC=3.3V; AVCC=3.3V analog; AGND linked to GND by R3.
External crystal and application–specific PWM auxiliary net: pending.





MCU local decoupling: C200..204 one per VDD; C205 bulk.
C206..208 analog/reference decoupling. R200/C209 reset.
R201/R202 boot pulldowns. J200 SWD. Internal clock startup.
Place decouplers beside respective supply pins during PCB design.

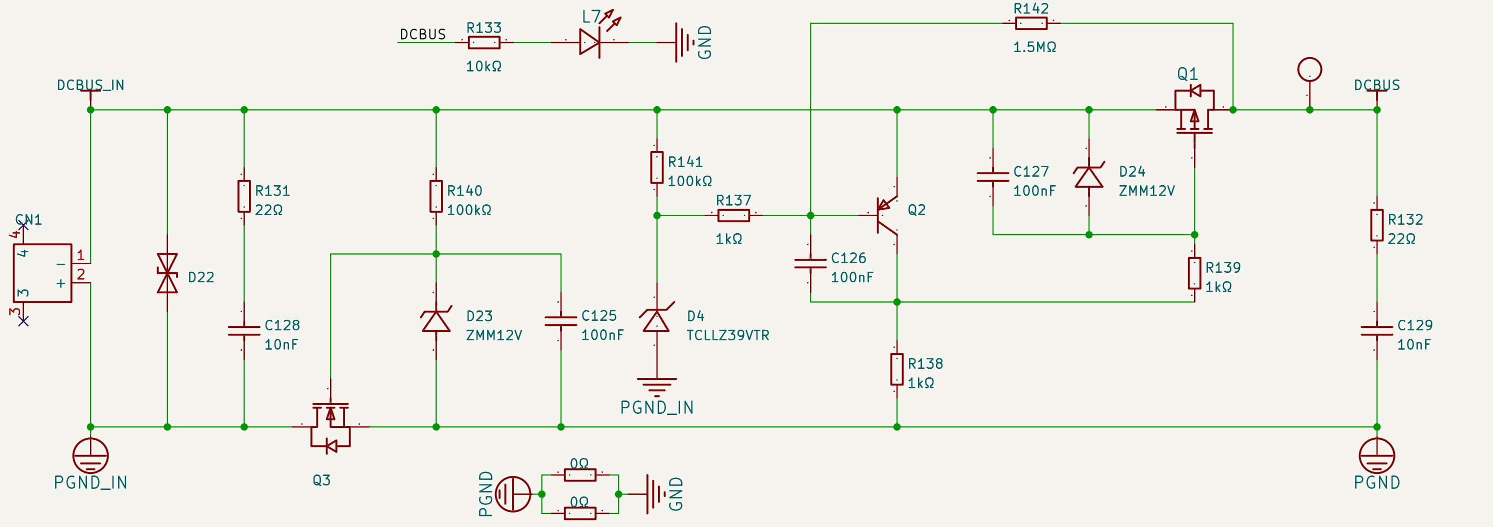
Alternate functions: PA8/9/10 AF1; PB13/14 AF2; PB15 AF1.
SPI0 PA4..7 AF0. TIMER2 PC6..8 AF0.
Reference: official GD32EB F505VxT7.xml and AFIO GD32F505.xml.

Sheet: /MCU/
File: MCU.kicad_sch

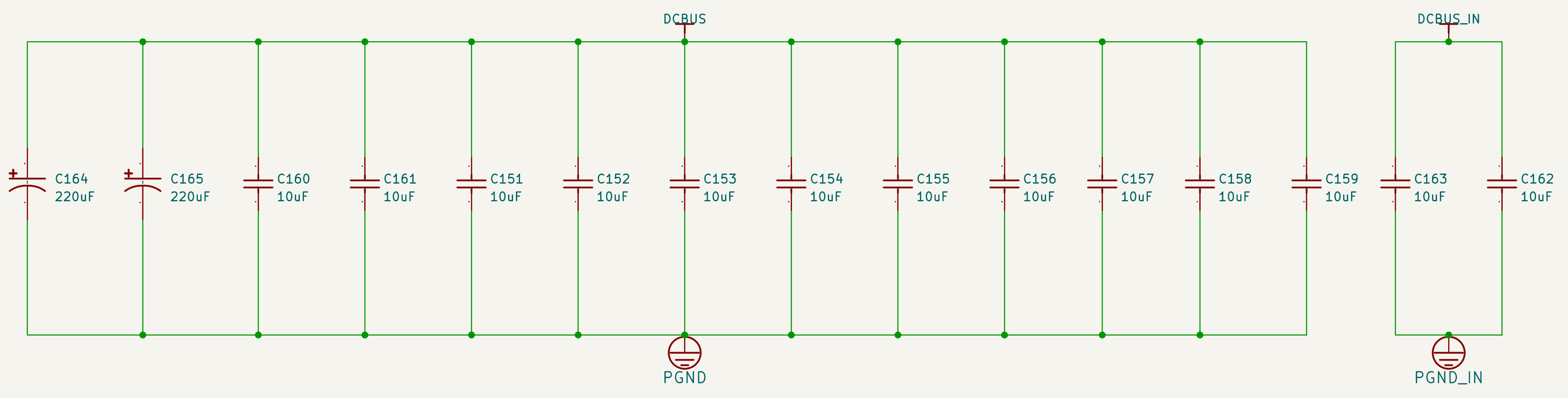
Title:

Size: A3
Date:
Rev:
KiCad E.D.A. 10.0.6–87–gae6d8f40c4
Id: 2/3

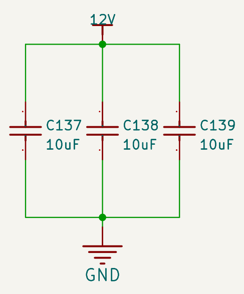
防反接&防过压&缓启动



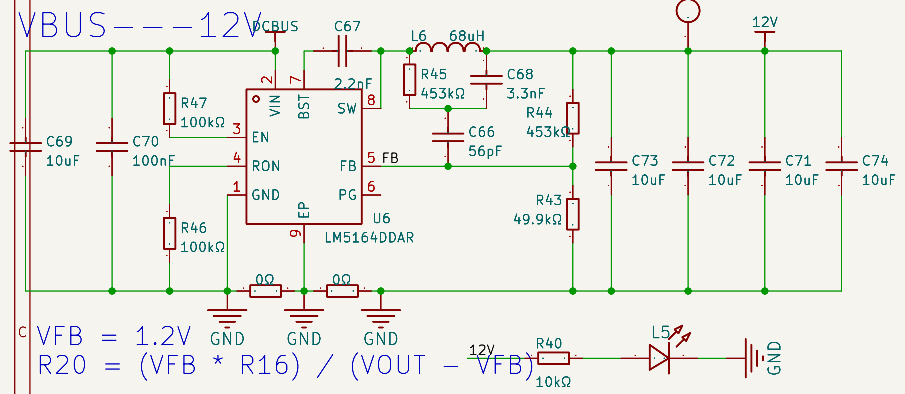
电源滤波



电源滤波



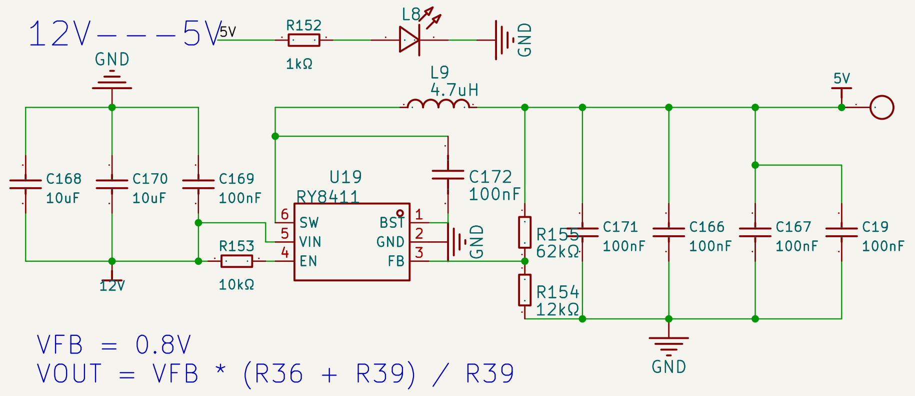
VBUS---12V



VFB = 1.2V

R20 = (VFB * R16) / (VOUT - VFB)

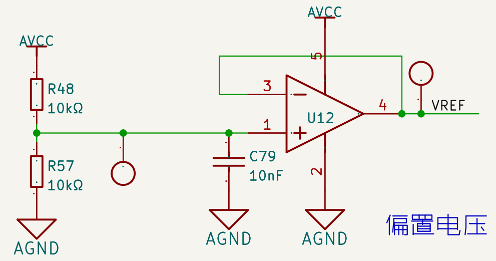
12V---5V



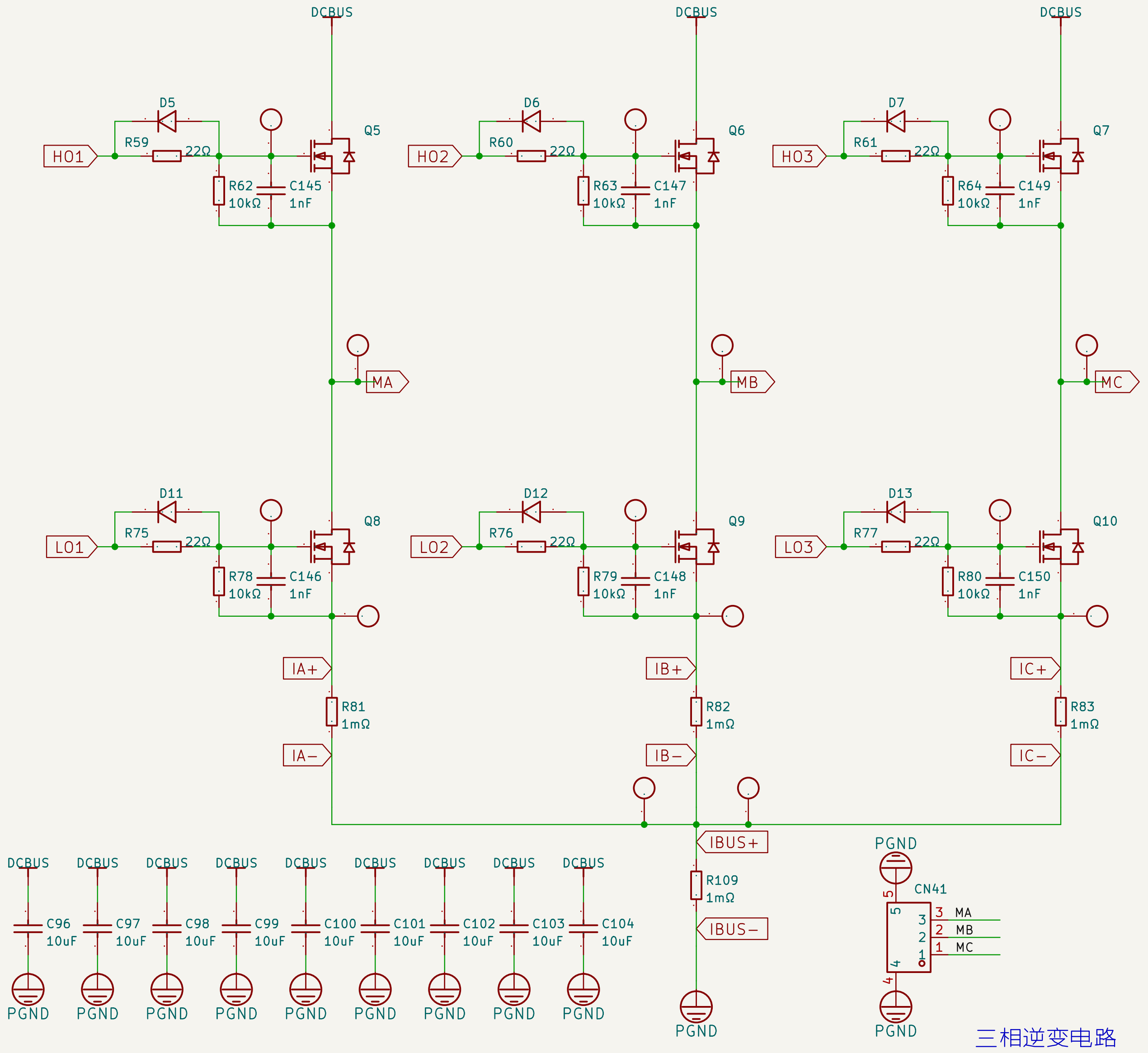
VFB = 0.8V

VOUT = VFB * (R36 + R39) / R39

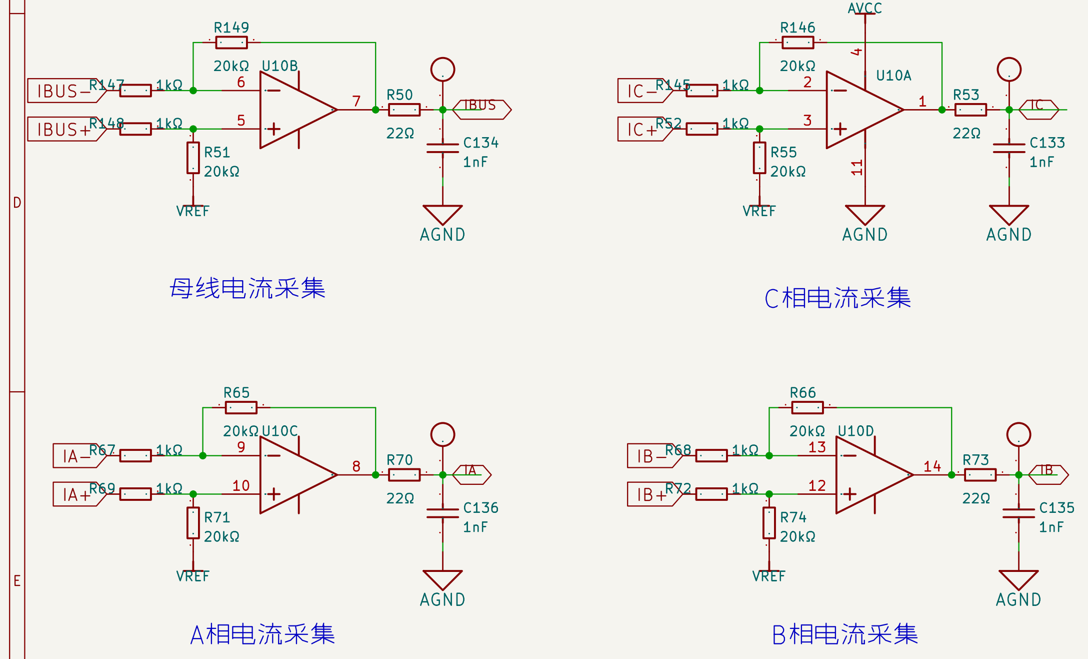
跟随器



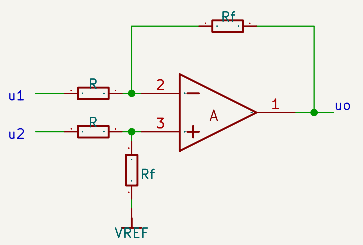
驱动电路



电流采样



电流采样计算



运放增益:k = Rf / R

偏置电压:VREF = 1.65V = 2048

采样电阻:Rm = 1mΩ

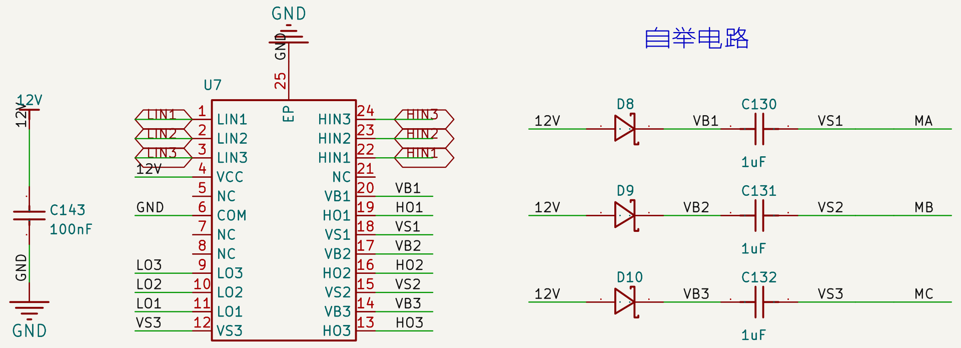
运放输出:uo = k * (u2 - u1)

u2 - u1 = I * Rm

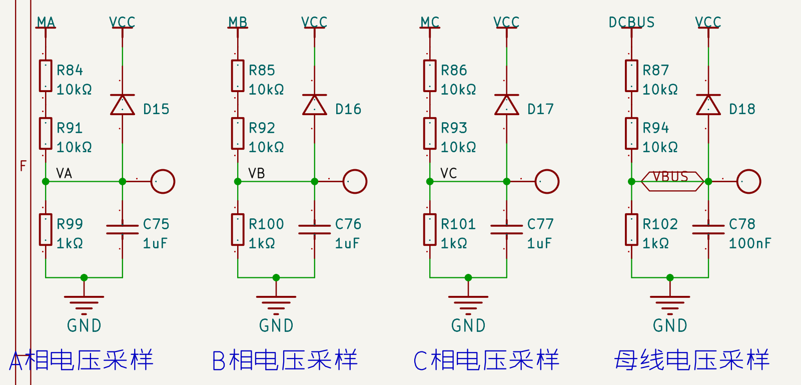
uo = (adc_val - 2048) * 3.3V/4096

采集电流:I = uo / (Rm * k)

三相半桥驱动



电压采样



采样点最高电压为: 3.3V + 二极管管压降

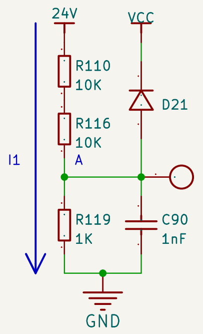
采样点电压应接近3.3V的一半,分辨率高，易采样

I1电流: 24V/(10k+10k+1k) = 1.14mA

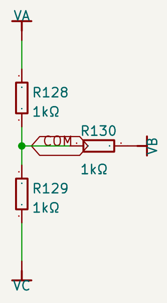
A点电压: I1 * 1k = 1.14V

缩小电压系数: k = (10k+10k+1k) / 1k

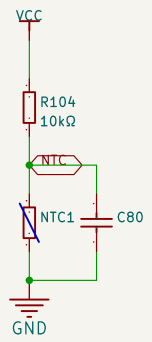
VA = adc_val * (3.3V/4096) * k



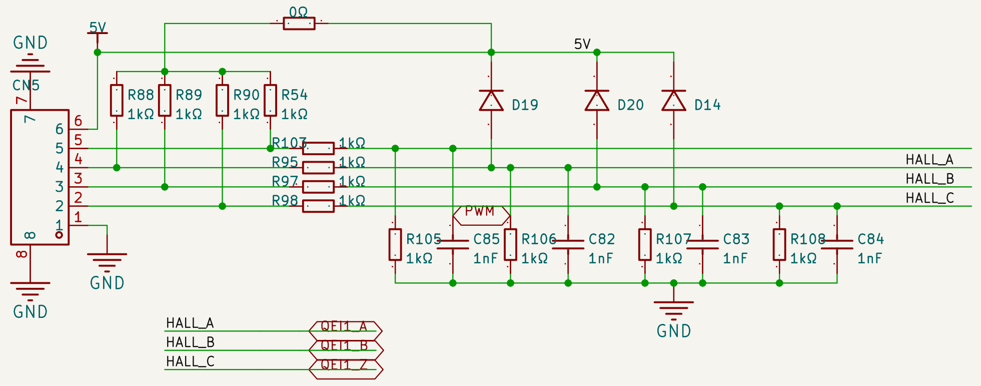
中性点采样



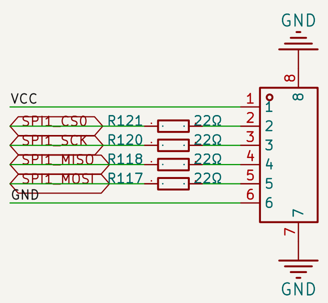
温度采样



霍尔接口



SPI接口



REFERENCE NOTES: U11/R56/R58/R143/R144 are an illustrative op-amp example.
CN1 source pin names conflict with rail polarity: verify connector before PCB.
10 imported Short-Symbol graphics replaced by electrical wires.
24V and DCBUS remain distinct source nets; auxiliary PWM is pending.

原理图	Schematic3	更新日期	2026-01-16
		创建日期	2026-01-16
图页	P2	物料编码	
绘制			
审阅			

Sheet: /FOC/
File: power.kicad_sch
工业嵌入式EtherCAT/HPM5E开发板_copy
Title:
Size: User　Date:　**Rev:**
KiCad E.D.A. 10.0.6-87-gae6d8f40c4　Id: 3/3

版本	尺寸	页	2	共	2
V1.0	A2	嘉立创EDA			